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Full Legal Name: Su Tint Tal

Institution: IIP International School

Category: Junior

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Submission Date: September 19, 2026

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Research Topic: Pharmacognosy, oncology and molecular biology

Title: Insect toxins compounds as oncological agents: a comparative analysis of cytotoxic mechanisms

Animals living in the wild are known to survive harsh environment where it is easy to spread diseases or catch diseases. However, they treat themselves through zoopharmacognosy (multidisciplinary process of the self-medication of animals). An example of this behavior is usually observed in chimpanzees (*Pan troglodytes*) and capuchins (*Cebus/Sapajus*); they are known for intentionally crushing venomous ants (*Camponotus spp*.) or millipedes [1][2], which has formic acid and venomous peptides, and rub them against their skin or fur to repel against painful mosquito bites, lethal ectoparasites, as well as prevent bacterial infections [2].

If these animals can herness these aggressive and toxic compounds of venomics for protection and survival, modern medicine can decode their molecular structures [3] to target cellular threats such as malignant neoplasm. For instance; bioactive toxins, especially cationic antimicrobial peptides (AMPs) such as melittin from honeybee (*Apis mellifera*) toxin and mastoparan found in wasp (*Vespa*) venom, have evolved to disrupt cellular membranes. Thus, malignant cells which are characterized by an excess amount of negatively charged molecules, they attract the positively charged insect toxins when the peptides are reprogrammed.

Research Problem

Malignant glioblastoma cells remain highly life-threatening due to the blood-brain barrier [4], also known as BBB: a highly selective border of cells that stops harmful substances in the blood from entering the brain, which blocks over ninety-eight percent of standard neuro-pharmaceuticals from reaching tumors. While nature demonstrates the defensive protection of insect chemistry through zoopharmacognosy (such as primates using venomous insects for self-medication) [2], applying these wild toxins into human oncology is limited by their profound lack of cellular selectivity [5]. In crude venom matrices, the therapeutic peptides melittin and mastoparan cause severe toxicity and lethal hemolysis in healthy blood cells while destroying tumors [6]. As a result, a critical gap exists in developing refined chemical extraction and purification protocols capable of isolating these toxin components to neutralize their toxicity while preserving their selective ability to trigger programmed cell death in malignant glioma tissues.

Research Questions

1. To what extent does harvest insect toxin components, specifically melittin and mastoparan demonstrate selective cytotoxicity against human glioblastoma cells while minimizing damage to healthy astrocytoma cells

2. How can specific chemical extraction protocols be enhanced to yield the highest purity and concentration of melittin and mastoparan from crude venom matrices?

3. Why do certain insect-derived toxin components demonstrate higher anti-proliferative efficiency in opposition to malignant cells when compared to standard chemotherapeutic **drugs?**

Hypothesis

At optimized, low nanomolar concentrations, the cationic peptides will selectively bind to the negatively charged phosphatidylserine glioma membranes, triggering cell death.

Literature Review

The exploration of natural economic biological resources for oncological therapies increasingly depends on zoopharmacognosy: the process by which non-human animals self-medicate [1]. Within natural ecosystems, primates like capuchin monkeys (*Cebus/Sapajus*) and chimpanzees (*Pan troglodytes*) provide observational data of the therapeutic value of insect chemistry. Primates engage in intentionally crushing and rubbing specific ants (*Camponotus spp.*) or toxic millipedes against their fur to detoxify formic acid and defensive peptides as a shield against pathogens [2]. Similarly, chimpanzees have been observed applying crushed insect pastes directly on open wounds to increase the rate of healing [9]. These behaviors demonstrate that insect secretions contain adaptively refined bio-active agents capable of cellular survival. If modern pharmacology can decode these defense systems, it can be used to target internal human cellular threats like cancer [10].

To transition these discoveries into human oncology, the research must maintain a precise distinction between crude venoms and isolated toxin components. Crude venom is a complex, macroscopic biological matrix containing volatile histamines, hyper-allergenic proteins and salts optimized as an evolutionary weapon to cause systemic tissue damage and lethal hemolysis in healthy blood cells [11]. In contrast, a toxin is a substance created by germs, plants and animals that are poisonous to humans [7]. Within Hymenoptera, the true biologic are low-molecular-weight peptide toxins, such as melittin and mastoparan. As raw venom matrices cause severe off-target toxicity; therefore, clinical translation relies entirely on purified or extracted toxin components to isolate these therapeutic peptides away from dangerous, allergy-inducing elements [10].

This molecular improvement is important when targeting malignant glioblastoma cells which remain resistant to standard therapies [17] due to Blood-Brain Barrier blocking over ninety-eight percent of small-molecule chemotherapeutics [12][13]. Purified cationic peptides like melittin and mastoparan possess the structural stability to interact with cellular membranes. While healthy cells maintain a neutral outer charge that minimizes peptide binding, malignant glioblastoma cells present a net negative charge due to overexpressed phosphatidylserine [11][14]. Upon binding, these purified toxins trigger transmembrane pore formation, driving localized necrosis or activating the mitochondrial apoptosis pathway to execute programmed cell death [15].

This research aims to improve extraction protocols to establish the precise therapeutic index, needed to transition these dangerous biological defenses into targeted neuro-oncological agents.

Methodology

This study adopts a quantitative methodology to investigate the extraction optimization, cytotoxicity and hemolytic indices of certain purified insect toxins components, such as melittin and mastoparan, against human glioma cell lines.

1. **High-performance liquid chromatography fractionation:** Crude venom matrices from Apis mellifera and Vespa species will be dissolved in deionized water and be injected into a High-performance liquid chromatography (HPLC) system using a C18 reverse phase column (a non-polar chromatographic tool packed with silica beads bonded to 18-carbon chains) [8]. Conduct two different phases as crude insect venom is filled with a significant number of different molecules that have completely separate chemical properties. A linear gradient of a mobile phase A and mobile phase B will be metered at a constant flow rate of 1.0 mL/min.

2. **Yield quantification:** Extracted peaks corresponding to melittin and mastoparan will be monitored through UV absorbance. Total protein concentration and percentage purity will be calculated using peak area integration software, namely OriginLab Origin, relative to known standard curves and synthetic peptides.

3. **Experimental Groups:** Human glioblastoma cell lines: U-84 MG and U-251 MG; and non-cancerous control cells: healthy human astrocytes; are seeded into 96-well plates at a fixed density. Cells will be treated with six log dilution concentrations (0.1 nM to 10 μM) of the purified toxins alongside a positive control and a negative control.

4. **Inhibitory concentration and Determination:** Following incubation periods of 24 hours, 48 hours and 72 hours; cell viability will be qualified colorimetrically using a CCK-8 or MTT assay. Optical density should be measured at 450 nm using a microplate reader. Cell survival percentages will then be plotted against log concentration values to calculate the half-maximum IC_{50} through non-linear regression analysis.

5. **Hemolysis assay:** Healthy human erythrocytes should be isolated through centrifugation and diluted to a 2% V/v suspension in Phosphate-buffered Saline (PBS). The red blood cell will be exposed to the identical concentration of purified toxins.

6. **Absorbance measurement:** After an hour incubation at 37 C, samples will be spun rapidly in a machine and the supernatant absorbance will be measured at 540 nm to quantify released hemoglobin. Total hemolysis will be established using one percent

Triton X-100 while gamma-hemolysis will be established using PBS. The Hemolytic Index will be calculated to create the Selectivity Index (SI = HC_{50} / IC_{50})

7. **Data verification:** All assays will be performed in a triplicate across three independent biological replicates to ensure statistical validity.

Limitation

The most critical limitation to the clinical translation of Hymenoptera-derived peptides is their inherent immunogenicity and the severe risk of Immunoglobulin E medicated allergic reactions [10], including systemic anaphylaxis. While laboratory isolation processes completely separate therapeutic peptides—melittin and mastoparan, from hyper-allergenic venom enzymes such like Phospholipase A2. However, these isolated toxins can still trigger a localized or systemic immune response in sensitive human subjects by stimulating mast cell degranulation and histamine release [16][15]. As a significant portion of the global population exhibits pre-existing hypersensitivity to bees and wasps, any future therapeutic application requires precise molecular modifications to shield the bio-active peptide from immune recognition while preserving its selective ability to destroy malignant glioma cells [17].

Expected Outcomes

According to the quantitative experimental design and hypothesis, this study expects to achieve the optimization of reverse phase HPLC and mobile phase gradients to yield isolated fractions of melittin and mastoparan with a verified chemical purity exceeding ninety-five percent, completely free of high molecular weight allergenic enzymes. When applied to 96-well plates, these purified toxin components are expected to generate ($p < 0.05$) sigmoidal dose response curves that establish clear IC_{50} value thresholds for inhibition of U-87 MG and U-251 MG human glioblastoma cell lines within 48-to-72-hour period. Additionally, the hemolysis assays are anticipated to show that these cationic peptides especially target the negatively charged phosphatidylserine on tumor membranes, proving a highly favorable Selectivity Index where healthy erythrocytes and normal astrocytes remain viable at therapeutic doses.

Conclusion

In conclusion, by decoding the natural defense systems observed in primate zoopharmacognosy, this study transitions raw biological weapons into precise pharmaceutical tools. Utilizing a demanding quantitative paradigm to isolate purified toxin components like melittin and mastoparan offers a visible path to circumvent traditional chemotherapeutic obstruction and selectively destroy malignant glioma membranes. Finally, this research hopes to provide the baseline of metrics necessary to neutralize the systemic toxicity of insect venoms, transforming them into safe, highly targeted and next-generation neuro-oncological therapeutics.

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